Conserved Totals in a Four-State Enzyme Kinetics Model

E. Kinetics

October 2026

Abstract

We track a single-substrate enzyme reaction through its four state variables: substrate S, complex C, free enzyme E and product P. Two totals are conserved along every trajectory — the enzyme total and the substrate total — and the product rises monotonically toward the initial substrate level.

1 The reaction scheme

With association rate κ, dissociation rate λ and catalytic rate μ, the mass-action dynamics read

$$\dot{S} = -\kappa S E + \lambda C, \tag{1}$$

$$\dot{C} = \kappa S E - \lambda C - \mu C, \tag{2}$$

$$\dot{E} = -\kappa S E + \lambda C + \mu C, \tag{3}$$

$$\dot{P} = \mu C. \tag{4}$$

2 Initial state and rates

The assay starts with substrate and enzyme only, at the recorded initial state $S(0) = 1$, $C(0) = 0$, $E(0) = 0.1$ and $P(0) = 0$. The nominal rates are $\kappa = 1.0$, $\lambda = 0.2$ and $\mu = 0.4$.

3 Discussion

Because the enzyme is either free or bound, the total $E + C$ stays constant; similarly $S + C + P$ never changes. The conservation of both totals is the internal check used for every reported trajectory.